

Case Study 2

AKSTA Statistical Computing

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2026-04-27

Table 1: You can use and extend this table to specify your AI usage (if used at all).

AI usage	Task a	Task b	Task c	Task d	Task e	Task f	Task g	Task h	Task i
<i>Correcting faulty code</i>	medium	medium	medium	medium	medium	medium	medium	medium	medium
<i>Improving existing code</i>	no	no	no	no	no	no	no	no	no
<i>Generating new code</i>	no	no	no	no	no	no	no	no	no
<i>Layout and formatting</i>	no	no	no	no	no	no	no	no	no
<i>Text correction and rewording</i>	heavy	heavy	heavy	heavy	heavy	heavy	heavy	heavy	heavy
...									

The CIA World Factbook provides intelligence on various aspects of 266 world entities, including history, people, government, economy, energy, geography, environment, communications, transportation, military, terrorism, and transnational issues. This case study involves analyzing world data from 2020, focusing on:

- **Education Expenditure (% of GDP)**
- **Youth Unemployment Rate (15-24 years)**
- **Net Migration Rate** (difference between the number of people entering and leaving a country per 1,000 persons)

The data was sourced from the CIA World Factbook Archives. You are required to use `dplyr` for data manipulation, while any package can be used for importing data.

Setup

```
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(stringr)
library(readxl)
```

Tasks:

a. Data Import and Cleaning

Load the following datasets from TUWEL and ensure that missing values are handled correctly and column names are clear. Each dataset should ultimately contain only two columns: **country** and the respective variable. Note that some data sets also contain information on the year when the value was last updated.

- `rawdata_369.txt` which contains the (estimated) public expenditure on education as a percent of GDP. *Pay attention! The delimiter is 2 or more white spaces (one space would not work as it would separate country names which contain a space); you have to skip the first two lines.*
- `rawdata_373.csv` which contains the (estimated) youth unemployment rate (15-24) per country
- `rawdata_347.txt` which contains (estimated) net migration rate per country.

Answer First of all we want to read all the data. According to the task each data set should ultimately contain only two columns: **country** and the respective measured variable, so the rank and the year of the last update are dropped during import.

```
# read data
raw_data_347 <- read_fwf(
  "rawdata_347.txt",
  col_positions = fwf_widths(c(8, 60, 10, 17),
                              c("rank", "country", "mig_rate", "year")),
  skip = 2,
  na = c("", "NA", "N/A")
) %>%
  select(country, mig_rate)

raw_data_369 <- read_fwf(
  "rawdata_369.txt",
  col_positions = fwf_widths(c(8, 60, 10, 12),
                              c("rank", "country", "edu_exp", "year")),
  skip = 2,
  na = c("", "NA", "N/A")
) %>%
  select(country, edu_exp)

raw_data_373 <- read_csv("rawdata_373.csv") %>%
  rename(country = country_name,
         youth_unempl_rate = youth_unempl_rate)

# show structure of read data
str(raw_data_347)

## tibble [227 x 2] (S3: tbl_df/tbl/data.frame)
## $ country : chr [1:227] "Syria" "British Virgin Islands" "Luxembourg" "Cayman Islands" ...
## $ mig_rate: num [1:227] 27.1 15.5 13.3 13 11.8 11.1 10.6 8.9 8.4 8.3 ...
str(raw_data_369)

## tibble [170 x 2] (S3: tbl_df/tbl/data.frame)
## $ country: chr [1:170] "Cuba" "Micronesia, Federated States of" "Solomon Islands" "Montserrat" ...
## $ edu_exp: num [1:170] 12.8 12.5 9.9 8.8 7.9 7.8 7.7 7.7 7.6 7.6 ...
str(raw_data_373)

## spc_tbl_ [181 x 2] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
```

```
## $ country          : chr [1:181] "French Polynesia" "Kosovo" "South Africa" "Libya" ...
## $ youth_unempl_rate: num [1:181] 56.7 55.4 53.4 48.7 47.1 46.2 45.4 42.2 42.2 40.5 ...
## - attr(*, "spec")=
## .. cols(
## ..   country_name = col_character(),
## ..   youth_unempl_rate = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

The two .txt files were parsed with `read_fwf()` from the `readr` package, which can be used to parse fixed-width-files. To do this properly we have to pass the widths of the columns in our files to `read_fwf`. Because the first two rows in our files don't contain any relevant data we use the `skip` parameter to avoid parsing them. The .csv file was read with `read_csv()`. Country names in that file contain a long block of trailing whitespaces, which can be safely dropped. This is being handled automatically because the parameter `trim_ws`, which tells our function to trim all leading and trailing whitespaces before parsing, is set to `TRUE` by default. In all three data sets we rename the country column to `country` so that the merging step in part b can use a single, common key.

```
# check for missing values
cat("raw_data_347 contains", sum(is.na(raw_data_347)), "missing values.\n")
```

```
## raw_data_347 contains 0 missing values.
```

```
cat("raw_data_369 contains", sum(is.na(raw_data_369)), "missing values.\n")
```

```
## raw_data_369 contains 0 missing values.
```

```
cat("raw_data_373 contains", sum(is.na(raw_data_373)), "missing values.\n")
```

```
## raw_data_373 contains 0 missing values.
```

After the import none of the three tibbles contains missing values, so no further NA handling is required at this point. (Missing values will appear naturally in part b when we take the union of all countries through a full join.)

b. Merging Raw Data

Merge the datasets using `dplyr` on a unique key and retain the union of all observations.

- What key are you using for merging?
- Return the dimensions of the merged dataset.

Answer

The only variable that appears in all three raw data sets is the country name, so it is the natural choice for merging. To retain the union of all observations (i.e. every country that appears in at least one of the three files) we use `full_join()`.

```
# merge data
merged_data <- raw_data_347 %>%
  full_join(raw_data_369, by = "country") %>%
  full_join(raw_data_373, by = "country")

# show data, it's structure and dimensions
merged_data
```

```
## # A tibble: 227 x 4
##   country          mig_rate edu_exp youth_unempl_rate
##   <chr>            <dbl>   <dbl>         <dbl>
## 1 Syria           27.1    NA             35.8
```

```
## 2 British Virgin Islands      15.5      2.5      NA
## 3 Luxembourg                  13.3      3.6     14.2
## 4 Cayman Islands              13        NA     13.8
## 5 Singapore                  11.8      2.9      9.1
## 6 Anguilla                    11.1      NA      NA
## 7 Bahrain                     10.6      2.3      5.3
## 8 Turks and Caicos Islands    8.9      2.9      NA
## 9 Aruba                       8.4      5.5      NA
## 10 Monaco                     8.3      1.5     26.6
## # i 217 more rows
```

```
str(merged_data)
```

```
## tibble [227 x 4] (S3: tbl_df/tbl/data.frame)
## $ country      : chr [1:227] "Syria" "British Virgin Islands" "Luxembourg" "Cayman Islands" ...
## $ mig_rate      : num [1:227] 27.1 15.5 13.3 13 11.8 11.1 10.6 8.9 8.4 8.3 ...
## $ edu_exp       : num [1:227] NA 2.5 3.6 NA 2.9 NA 2.3 2.9 5.5 1.5 ...
## $ youth_unempl_rate: num [1:227] 35.8 NA 14.2 13.8 9.1 NA 5.3 NA NA 26.6 ...
```

```
dim(merged_data)
```

```
## [1] 227  4
```

```
# count NA values
```

```
apply(is.na(merged_data), MARGIN=2, FUN=sum)
```

```
##           country      mig_rate      edu_exp youth_unempl_rate
##           0           0           57           46
```

The merged dataset has 227 rows (countries) and 4 columns: the country name plus the three measured variables. Countries that are missing from one of the original files now have NA for the corresponding variable. For now this is fine, but one has to keep this fact in mind when doing later analyses.

c. Enriching Data with Income Classification

Obtain country income classification (low, lower-middle, upper-middle, high) from the World Bank and merge it with the dataset.

- Identify common variables between datasets. Can they be used for merging? Why or why not?
- Since ISO codes are standardized, download and use the CIA country data codes for merging. Make sure you are not losing any of the countries in your original data set when merging.

Answer

The only variable shared between our merged CIA data set and the World Bank income classification is the country name. In principle this column could be used as a merge key, but in practice it is not reliable: the two sources use slightly different spellings and conventions for several countries (for example *Macedonia* vs. *North Macedonia*, *Turkey* vs. *Turkey (Turkiye)* etc.). A name-based join would silently drop every such country. Standardised ISO-3 codes do not have this problem, so we use the CIA *Country Data Codes* file to attach a GENC (ISO-3) code to every country in our data and then merge the World Bank classification on that code.

```
# read country codes from .csv
```

```
country_codes <- read_csv("Country Data Codes.csv") %>%
  select(Name, GENC)
```

Before joining we have to make the country names in `merged_data` agree with those in the codes table. A quick anti-join shows which names cannot be matched directly:

```
# check if there are some mismatches
merged_data %>%
  anti_join(country_codes, by = c("country" = "Name")) %>%
  pull(country)
```

```
## [1] "Macedonia" "Turkey"
```

Only two countries are affected: *Macedonia* and *Turkey*. After looking them up manually in `country_codes.csv` we found out that this is simple name mismatch. *Macedonia* is called *North Macedonia* in the codes file and *Turkey* is called *Turkey (Turkiye)*. We rename them so that no country is lost in the join.

```
# rename Macedonia and Turkey
merged_data <- merged_data %>%
  mutate(country = recode(country,
                           "Macedonia" = "North Macedonia",
                           "Turkey"    = "Turkey (Turkiye)"))
```

```
# merge data with country_codes
merged_data <- merged_data %>%
  left_join(country_codes, by = c("country" = "Name"))
```

```
# check if all countries have a country_code
stopifnot(all(!is.na(merged_data$GENC)))
```

```
# show data, structure and dimensions
merged_data
```

```
## # A tibble: 227 x 5
##   country          mig_rate edu_exp youth_unempl_rate GENC
##   <chr>            <dbl>   <dbl>             <dbl> <chr>
## 1 Syria            27.1    NA              35.8 SYR
## 2 British Virgin Islands 15.5    2.5             NA   VGB
## 3 Luxembourg       13.3    3.6             14.2 LUX
## 4 Cayman Islands    13      NA              13.8 CYM
## 5 Singapore        11.8    2.9             9.1  SGP
## 6 Anguilla          11.1    NA              NA   AIA
## 7 Bahrain           10.6    2.3             5.3  BHR
## 8 Turks and Caicos Islands 8.9    2.9             NA   TCA
## 9 Aruba              8.4    5.5             NA   ABW
## 10 Monaco            8.3    1.5             26.6 MCO
## # i 217 more rows
```

```
str(merged_data)
```

```
## tibble [227 x 5] (S3: tbl_df/tbl/data.frame)
## $ country      : chr [1:227] "Syria" "British Virgin Islands" "Luxembourg" "Cayman Islands" ...
## $ mig_rate     : num [1:227] 27.1 15.5 13.3 13 11.8 11.1 10.6 8.9 8.4 8.3 ...
## $ edu_exp      : num [1:227] NA 2.5 3.6 NA 2.9 NA 2.3 2.9 5.5 1.5 ...
## $ youth_unempl_rate: num [1:227] 35.8 NA 14.2 13.8 9.1 NA 5.3 NA NA 26.6 ...
## $ GENC         : chr [1:227] "SYR" "VGB" "LUX" "CYM" ...
```

```
dim(merged_data)
```

```
## [1] 227 5
```

Every country in our data set now carries an ISO-3 code. Next we read the World Bank classification (sheet

List of economies of the file CLASS_2025_10_07.xlsx, which was pulled from the link in the instructions) and keep only the ISO-3 code and the income group. The four income classes are recoded to the short labels L, LM, UM, H and stored as an ordered factor so that subsequent tables and plots respect their ordering.

```
# read income classification data
wb_income_class <- read_excel("CLASS_2025_10_07.xlsx",
                              sheet = "List of economies") %>%

select(Code, `Income group`) %>%
rename(GENC = Code, income_group = `Income group`) %>%
mutate(income_status = recode(income_group,
                              "Low income" = "L",
                              "Lower middle income" = "LM",
                              "Upper middle income" = "UM",
                              "High income" = "H"),
       income_status = factor(income_status,
                              levels = c("L", "LM", "UM", "H"),
                              ordered = TRUE)) %>%

select(GENC, income_status)

# merge incomeclass with our data using country codes
merged_data <- merged_data %>%
  left_join(wb_income_class, by = "GENC")

# show data, structure and dimensions
merged_data
```

```
## # A tibble: 227 x 6
##   country          mig_rate edu_exp youth_unempl_rate GENC income_status
##   <chr>            <dbl>   <dbl>          <dbl> <chr>   <ord>
## 1 Syria            27.1    NA              35.8 SYR   L
## 2 British Virgin Islands 15.5    2.5             NA   VGB   H
## 3 Luxembourg        13.3    3.6             14.2 LUX   H
## 4 Cayman Islands     13      NA              13.8 CYM   H
## 5 Singapore         11.8    2.9             9.1  SGP   H
## 6 Anguilla           11.1    NA              NA   AIA   <NA>
## 7 Bahrain            10.6    2.3             5.3  BHR   H
## 8 Turks and Caicos Isla~ 8.9     2.9             NA   TCA   H
## 9 Aruba              8.4     5.5             NA   ABW   H
## 10 Monaco            8.3     1.5            26.6 MCO   H
## # i 217 more rows
```

```
str(merged_data)
```

```
## tibble [227 x 6] (S3: tbl_df/tbl/data.frame)
## $ country      : chr [1:227] "Syria" "British Virgin Islands" "Luxembourg" "Cayman Islands" ...
## $ mig_rate     : num [1:227] 27.1 15.5 13.3 13 11.8 11.1 10.6 8.9 8.4 8.3 ...
## $ edu_exp      : num [1:227] NA 2.5 3.6 NA 2.9 NA 2.3 2.9 5.5 1.5 ...
## $ youth_unempl_rate: num [1:227] 35.8 NA 14.2 13.8 9.1 NA 5.3 NA NA 26.6 ...
## $ GENC         : chr [1:227] "SYR" "VGB" "LUX" "CYM" ...
## $ income_status : Ord.factor w/ 4 levels "L"<"LM"<"UM"<...: 1 4 4 4 4 NA 4 4 4 4 ...
```

```
dim(merged_data)
```

```
## [1] 227 6
```

The use of `left_join()` guarantees that no country from the original data set is dropped. For some countries the World Bank has no classification (a handful of small territories such as *Anguilla* or *Kosovo*) keep NA in

the new `income_status` column:

```
merged_data %>%  
  filter(is.na(income_status)) %>%  
  pull(country)
```

```
## [1] "Anguilla"  
## [2] "Jersey"  
## [3] "Guernsey"  
## [4] "Montserrat"  
## [5] "Saint Helena, Ascension, and Tristan da Cunha"  
## [6] "Ethiopia"  
## [7] "Kosovo"  
## [8] "Saint Barthelemy"  
## [9] "Venezuela"  
## [10] "West Bank"  
## [11] "Wallis and Futuna"  
## [12] "Gaza Strip"  
## [13] "Saint Pierre and Miquelon"  
## [14] "Cook Islands"
```

d. Adding Geographical Information

Introduce continent and subcontinent (or region) data for each country.

- Find and download an appropriate online resource.
- Merge this information into the dataset, naming the final dataset `df_vars`. Make sure you are not losing any of the countries in your original data set when merging.

We found this source <https://unstats.un.org/unsd/methodology/m49/overview/>, which satisfies our need to introduce continent and subcontinent data for each country.

```
geographical_info <- read.csv(  
  "unsd.csv",  
  header = TRUE,  
  sep = ";"  
)  
  
# removing redundant country column  
geographical_info$Global.Code <- NULL  
geographical_info$Global.Name <- NULL  
geographical_info$Region.Code <- NULL  
geographical_info$Global.Code <- NULL  
geographical_info$Sub.region.Code <- NULL  
geographical_info$Intermediate.Region.Code <- NULL  
geographical_info$Intermediate.Region.Name <- NULL  
geographical_info$Country.or.Area <- NULL  
geographical_info$M49.Code <- NULL  
geographical_info$ISO.alpha2.Code <- NULL  
geographical_info$Least.Developed.Countries..LDC. <- NULL  
geographical_info$Land.Locked.Developing.Countries..LLDC. <- NULL  
geographical_info$Small.Island.Developing.States..SIDS. <- NULL  
  
# renaming columns to have a coherent naming  
colnames(geographical_info) <- c("continent", "subcontinent", "iso-code")
```

```
head(geographical_info)
```

```
##   continent    subcontinent iso-code
## 1   Africa Northern Africa      DZA
## 2   Africa Northern Africa      EGY
## 3   Africa Northern Africa      LBY
## 4   Africa Northern Africa      MAR
## 5   Africa Northern Africa      SDN
## 6   Africa Northern Africa      TUN
```

We are using a left join here so that we don't lose any countries from the original dataset.

```
df_vars <- merged_data %>%
  left_join(geographical_info, by= c("GENC" = "iso-code"))
head(df_vars)
```

```
## # A tibble: 6 x 8
##   country      mig_rate edu_exp youth_unempl_rate GENC income_status continent
##   <chr>          <dbl>   <dbl>          <dbl> <chr> <ord>      <chr>
## 1 Syria          27.1     NA              35.8 SYR  L        Asia
## 2 British Virg~  15.5     2.5              NA   VGB  H        Americas
## 3 Luxembourg     13.3     3.6              14.2 LUX  H        Europe
## 4 Cayman Islan~  13        NA              13.8 CYM  H        Americas
## 5 Singapore     11.8     2.9              9.1  SGP  H        Asia
## 6 Anguilla       11.1     NA              NA   AIA  <NA>     Americas
## # i 1 more variable: subcontinent <chr>
```

e. Data Tidiness and Summary Statistics

- Evaluate the tidiness of `df_vars` (observational units, variables, fixed vs. measured variables). Make adjustments to tidy the data, if necessary.

The dataset currently adheres to the standards of tidy data:

- Every column represents a single variable, and every row represents a unique observation without duplicates.
- Fixed variables include: `country`, `GENC`, `income_status`, `continent`, and `subcontinent`.
- Measured variables include: `mig_rate`, `edu_exp`, and `youth_unempl_rate`

Overall, the data is already tidy, but a few minor tweaks can be made to improve readability:

- The `GENC` column name isn't immediately clear, renaming it to `iso_code` would be more intuitive.
- The columns should be reordered so that all fixed variables are grouped together, followed by the measured variables.

```
# Rename GENC to iso-code
df_vars <- df_vars %>% rename(
  iso_code = GENC
)

# sort columns
df_vars <- df_vars %>%
  select("country", "iso_code", "income_status", "continent", "subcontinent", "mig_rate", "edu_exp", "youth_unempl_rate")

head(df_vars)
```



```
## # A tibble: 6 x 8
##   country      iso_code income_status continent subcontinent mig_rate edu_exp
##   <chr>        <chr>    <ord>        <chr>    <chr>        <dbl>   <dbl>
## 1 Syria        SYR      L           Asia     Western Asia   27.1    NA
## 2 British Virgin~ VGB      H           Americas Latin Ameri~   15.5    2.5
## 3 Luxembourg    LUX      H           Europe   Western Eur~   13.3    3.6
## 4 Cayman Islands CYM      H           Americas Latin Ameri~   13      NA
## 5 Singapore     SGP      H           Asia     South-easte~   11.8    2.9
## 6 Anguilla      AIA      <NA>        Americas Latin Ameri~   11.1    NA
## # i 1 more variable: youth_unempl_rate <dbl>
```

- Create a frequency table for the income status variable and briefly interpret the results.

The dataset is skewed toward wealthier categories, with over 60% of the observations falling into the Upper Middle and High-Income brackets. Conversely, the Low-Income bracket is underrepresented. This imbalance must be considered when deriving results or drawing conclusions from the data, as it may introduce bias.

```
income_freq <- function () {df_vars %>%
  count(income_status, name = "n") %>%
  mutate(percent = round(100 * n / sum(n), 1))
}
head(income_freq())
```

```
## # A tibble: 5 x 3
##   income_status      n percent
##   <ord>          <int>   <dbl>
## 1 L              25     11
## 2 LM             49    21.6
## 3 UM             53    23.3
## 4 H              86    37.9
## 5 <NA>           14     6.2
```

- Analyze the distribution of income status across continents by computing absolute and relative frequencies. Comment on the findings.

We can see a quite expected overall picture of how the income groups are distributed around the world: In Africa, most countries are classified as lower middle or low income countries, however there are a few countries with upper middle income and even one with high income. Europe leads with over 80% high income countries. Asia has many countries in the lower middle field but also some countries with higher income. Oceania has more high income countries (~43%) but also 28% classified as low middle income.

```
library(tidyr)

income_by_continent <- df_vars %>%
  group_by(continent, income_status) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(continent) %>%
  mutate(percent = round(100 * n / sum(n), 1)) %>%
  ungroup() %>%
  arrange(continent, desc(percent))
# easier to analyze
income_by_continent_wide <- income_by_continent %>%
  pivot_wider(names_from = income_status,
              values_from = c(n, percent),
              values_fill = 0)

income_by_continent_wide
```

```
## # A tibble: 6 x 11
##   continent  n_LM  n_L  n_UM  n_NA  n_H percent_LM percent_L percent_UM
##   <chr>      <int> <int> <int> <int> <int>      <dbl>      <dbl>      <dbl>
## 1 Africa      23    21     8     2     1      41.8      38.2      14.5
## 2 Americas     4     0    18     5    23       8       0       36
## 3 Asia       17     4    14     0    14      34.7      8.2      28.6
## 4 Europe       0     0     8     2    38       0       0      16.7
## 5 Oceania      5     0     5     2     9      23.8       0      23.8
## 6 <NA>         0     0     0     3     1       0       0       0
## # i 2 more variables: percent_NA <dbl>, percent_H <dbl>
```

- Using the distribution of income status across continents, identify which countries are the only ones in their income group across the continent. Discuss briefly.

Seychelles is the only high-income country in Africa, likely due to its thriving tourism sector

```
df_vars %>%
  filter(continent=="Africa", income_status == "H") %>%
  select(country)
```

```
## # A tibble: 1 x 1
##   country
##   <chr>
## 1 Seychelles
```

f. Further Summary Statistics and Insights

- Create a table of average (mean and median) values for expenditure, youth unemployment rate and net migration rate separated into income status. Make sure that in the output, the ordering of the income classes is proper (i.e., L, LM, UM, H or the other way around). Briefly comment the results and any differences between the mean and median.

```
summary_stats <- df_vars %>%
  group_by(income_status) %>%
  summarise(
    mean_expenditure = mean(edu_exp, na.rm = TRUE),
    median_expenditure = median(edu_exp, na.rm = TRUE),
    mean_unemployment = mean(youth_unempl_rate, na.rm = TRUE),
    median_unemployment = median(youth_unempl_rate, na.rm = TRUE),
    mean_migration = mean(mig_rate, na.rm = TRUE),
    median_migration = median(mig_rate, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(income_status)

summary_stats
```

```
## # A tibble: 5 x 7
##   income_status mean_expenditure median_expenditure mean_unemployment
##   <ord>          <dbl>          <dbl>          <dbl>
## 1 L              3.53              3.1            14.9
## 2 LM             4.43              3.9            15.8
## 3 UM             4.51              4.1            22.1
## 4 H              4.6               4.7            16.5
## 5 <NA>           5.76              5.3            35.9
## # i 3 more variables: median_unemployment <dbl>, mean_migration <dbl>,
## #   median_migration <dbl>
```

Education expenditure steadily increases with country wealth, while high income nations are the only group experiencing net positive migration. Youth unemployment is highest in upper middle income countries. Across most brackets, the mean youth unemployment rate is higher than the median, indicating that a few countries with severe youth unemployment are skewing the averages upwards. Conversely, middle income brackets show means substantially lower than their medians for migration, driven by extreme emigration in a handful of nations.

- Look at the standard deviation and the interquartile range of the variables per income status instead of the location statistics above. Do you gain additional insights? Briefly comment the results.

```
dispersion_stats <- df_vars %>%
  group_by(income_status) %>%
  summarise(
    sd_expenditure = sd(edu_exp, na.rm = TRUE),
    iqr_expenditure = IQR(edu_exp, na.rm = TRUE),
    sd_unemployment = sd(youth_unempl_rate, na.rm = TRUE),
    iqr_unemployment = IQR(youth_unempl_rate, na.rm = TRUE),
    sd_migration = sd(mig_rate, na.rm = TRUE),
    iqr_migration = IQR(mig_rate, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(income_status)

dispersion_stats

## # A tibble: 5 x 7
##   income_status sd_expenditure iqr_expenditure sd_unemployment iqr_unemployment
##   <ord>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 L             1.73           2.65           12.8           15.1
## 2 LM            2.24           2.55           11.7           14.4
## 3 UM            1.78           1.65           12.7           16.4
## 4 H             1.49           1.8            10.3           14.0
## 5 <NA>          1.73           0.600          16.0           17
## # i 2 more variables: sd_migration <dbl>, iqr_migration <dbl>
```

By looking at the spread, we learn that High income countries are generally more uniform in their policies (like education spending), whereas Lower and Middle income brackets are highly volatile, especially regarding migration, where the data is being heavily distorted by a few countries experiencing extreme population movements.

- Extend the analysis of the statistics median and IQR to **each income status and continent combination**. Play around with displaying the resulting table. Use `pivot_longer()` and/or `pivot_wider()` to generate different outputs. Discuss the results as well as the readability of the different tables.

```
# Creating a comprehensive table with median and IQR for each variable by income group and continent
combined_stats <- df_vars %>%
  group_by(continent, income_status) %>%
  summarise(
    median_expenditure = median(edu_exp, na.rm = TRUE),
    iqr_expenditure = IQR(edu_exp, na.rm = TRUE),
    median_unemployment = median(youth_unempl_rate, na.rm = TRUE),
    iqr_unemployment = IQR(youth_unempl_rate, na.rm = TRUE),
    median_migration = median(mig_rate, na.rm = TRUE),
    iqr_migration = IQR(mig_rate, na.rm = TRUE),
    n_countries = n(),
    .groups = "drop"
  ) %>%
```

```

arrange(continent, income_status)

head(combined_stats)

## # A tibble: 6 x 9
##   continent income_status median_expenditure iqr_expenditure median_unemployment
##   <chr>      <ord>                <dbl>          <dbl>          <dbl>
## 1 Africa    L                2.95            2.82            8.7
## 2 Africa    LM               3.55            2.05           16.5
## 3 Africa    UM               4.95            1.33           36
## 4 Africa    H                4.4             0             11.6
## 5 Africa    <NA>            4.7             0             25.2
## 6 Americas  LM               5.25            2.4            8.5
## # i 4 more variables: iqr_unemployment <dbl>, median_migration <dbl>,
## #   iqr_migration <dbl>, n_countries <int>

# Format 2: Focus on one variable at a time, income groups as columns
unemployment_by_continent_income <- combined_stats %>%
  select(continent, income_status, median_unemployment, iqr_unemployment, n_countries) %>%
  pivot_wider(
    id_cols = continent,
    names_from = income_status,
    values_from = c(median_unemployment, iqr_unemployment, n_countries),
    values_fill = NA
  )

# Display the unemployment focused table
unemployment_by_continent_income

## # A tibble: 6 x 16
##   continent median_unemployment_L median_unemployment_LM median_unemployment_UM
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 Africa                8.7            16.5           36
## 2 Americas              NA              8.5           14.5
## 3 Asia                 24.5            13           16.6
## 4 Europe                NA              NA            29.6
## 5 Oceania               NA             10.4           18
## 6 <NA>                  NA              NA            NA
## # i 12 more variables: median_unemployment_H <dbl>,
## #   median_unemployment_NA <dbl>, iqr_unemployment_L <dbl>,
## #   iqr_unemployment_LM <dbl>, iqr_unemployment_UM <dbl>,
## #   iqr_unemployment_H <dbl>, iqr_unemployment_NA <dbl>, n_countries_L <int>,
## #   n_countries_LM <int>, n_countries_UM <int>, n_countries_H <int>,
## #   n_countries_NA <int>

# Format 3: Focus on continents with variables and income groups nested
continent_focused <- combined_stats %>%
  select(-n_countries) %>%
  pivot_longer(
    cols = c(median_expenditure, iqr_expenditure, median_unemployment,
              iqr_unemployment, median_migration, iqr_migration),
    names_to = "statistic",
    values_to = "value"
  ) %>%
  mutate(

```

```

variable = case_when(
  grepl("expenditure", statistic) ~ "Education Expenditure",
  grepl("unemployment", statistic) ~ "Youth Unemployment",
  grepl("migration", statistic) ~ "Net Migration",
  TRUE ~ NA_character_
),
stat_type = ifelse(grepl("median", statistic), "Median", "IQR")
) %>%
pivot_wider(
  id_cols = c(continent, variable, income_status),
  names_from = stat_type,
  values_from = value
)

# Display the continent-focused table
head(continent_focused)

```

```

## # A tibble: 6 x 5
##   continent variable      income_status Median   IQR
##   <chr>      <chr>          <ord>      <dbl> <dbl>
## 1 Africa    Education Expenditure L          2.95  2.82
## 2 Africa    Youth Unemployment   L          8.7   12.4
## 3 Africa    Net Migration        L         -0.9   3.2
## 4 Africa    Education Expenditure LM         3.55  2.05
## 5 Africa    Youth Unemployment   LM        16.5  22.2
## 6 Africa    Net Migration        LM        -0.4   1.65

```

Breaking the data down by continent reveals wealth disparities. Africa is dominated by lower income nations, whereas Europe consists of upper middle and high income countries. Additionally, youth unemployment trends do not follow a single rule. Unemployment peaks in upper middle income nations in Africa and Europe, but it is highest in low income countries in Asia. In the Americas and Oceania, unemployment peaks in high income nations, highlighting data volatility.

When comparing table readability, the long, nested format (Format 3) is vastly superior for professional reporting. It neatly pairs the median and IQR while keeping the table narrow and visually clean. In contrast, the wide format (Format 2) is difficult to read on a screen or page due to horizontal bloat and a clutter of NA values for non-existent continent-income combinations, though it does remain useful for quick, left-to-right data exploration.

- Identify countries performing well in terms of both **youth unemployment** and **net migration rate** (top 25% in net migration and bottom 25% in youth unemployment within their continent).

```

# Calculate quartiles for each variable within continents
quartiles <- df_vars %>%
  group_by(continent) %>%
  mutate(
    unemployment_quantile = ntile(youth_unempl_rate, 4),
    migration_quantile = ntile(mig_rate, 4)
  ) %>%
  ungroup()

# Identify countries in bottom 25% for unemployment and top 25% for migration
well_performing_countries <- quartiles %>%
  filter(unemployment_quantile == 1, migration_quantile == 4) %>%
  select(country, continent, youth_unempl_rate, mig_rate, income_status) %>%
  arrange(continent, country)

```

```
# Display the well-performing countries
well_performing_countries
```

```
## # A tibble: 13 x 5
##   country          continent youth_unempl_rate mig_rate income_status
##   <chr>            <chr>          <dbl>      <dbl> <ord>
## 1 Benin            Africa             5.6        0.3 LM
## 2 Cote d'Ivoire     Africa             5.5        1.2 LM
## 3 Togo              Africa             3.9         0   L
## 4 United States     Americas           8.6         3   H
## 5 Bahrain           Asia              5.3       10.6 H
## 6 Macau             Asia              5.3         3.3 H
## 7 Qatar             Asia              0.4         6.5 H
## 8 United Arab Emirates Asia             6.9         7.6 H
## 9 Malta             Europe            9.1         6.6 H
## 10 Norway           Europe            9.7         4   H
## 11 Switzerland      Europe            7.9         4.6 H
## 12 Palau            Oceania           5.6         0.9 H
## 13 Papua New Guinea Oceania           3.6         0   LM
```

These identified countries perform well compared to their continental peers, with low youth unemployment and high net migration rates. This combination indicates economic opportunities that retain local youth in the workforce and attract migrants.

Interestingly, these countries span different income groups, suggesting that relative performance within a continent is not solely determined by income classification. The presence of countries from various regions demonstrates that success in these metrics can be achieved through different policy approaches adapted to local conditions.

Policymakers might look to these nations as case studies for successful strategies in youth employment and migration management within their respective continents.

g. Conditional Probabilities

Estimate the following based on the observed frequencies in the data:

- What is the (posterior or conditional) probability that a European country belongs to the high income group? What is the prior probability that a country belongs to the high income group?
- Given a country has high youth unemployment (above %25), what is the probability that it also has negative net migration?

Answer

```
# Prior probability that a country belongs to the high income group
p_high_income <- mean(df_vars$income_status == "H", na.rm = TRUE)

# Conditional probability that a European country belongs to the high income group
p_high_given_europe <- df_vars %>%
  filter(continent == "Europe") %>%
  summarize(prob = mean(income_status == "H", na.rm = TRUE)) %>%
  pull(prob)

# Given high youth unemployment (>25%), probability of negative net migration
p_neg_mig_given_high_unemp <- df_vars %>%
  filter(youth_unempl_rate > 25) %>%
  summarize(prob = mean(mig_rate < 0, na.rm = TRUE)) %>%
```

```
pull(prob)

cat("Prior probability of high income: ", round(p_high_income * 100, 2), "%\n")

## Prior probability of high income: 40.38 %

cat("Probability of high income given Europe: ", round(p_high_given_europe * 100, 2), "%\n")

## Probability of high income given Europe: 82.61 %

cat("Probability of negative migration given high unemployment (>25%): ",
    round(p_neg_mig_given_high_unemp * 100, 2), "%\n")

## Probability of negative migration given high unemployment (>25%): 61.22 %
```

The prior probability of a country belonging to the high-income group is computed by taking the proportion of all countries with an income status of “H”. The conditional probability of a European country being in the high-income group restricts the denominator to only European countries. As expected, the probability of being high income is significantly higher when we know the country is in Europe compared to the global baseline.

Furthermore, the conditional probability of experiencing negative net migration given a high youth unemployment rate (above 25%) shows that a large proportion of these countries suffer from “brain drain” or population loss as young people migrate to find better economic opportunities.

h. Simpson’s Paradox Analysis

Investigate whether an overall trend in youth unemployment rate in the high and low income groups reverses when analyzed at the continent level. E.g., does the youth unemployment rate appear lower in low-income countries overall, but higher when controlling for continent? Explain the results and possible reasons behind this paradox.

Answer

```
# Overall youth unemployment rate by income group (comparing L and H)
overall_unemp <- df_vars %>%
  filter(income_status %in% c("L", "H")) %>%
  group_by(income_status) %>%
  summarize(mean_unemp = mean(youth_unempl_rate, na.rm = TRUE),
            n = n(), .groups = "drop")

overall_unemp
```

```
## # A tibble: 2 x 3
##   income_status mean_unemp      n
##   <ord>          <dbl> <int>
## 1 L            14.9     25
## 2 H            16.5     86
```

```
# Continent-level youth unemployment rate by income group
continent_unemp <- df_vars %>%
  filter(income_status %in% c("L", "H")) %>%
  group_by(continent, income_status) %>%
  summarize(mean_unemp = mean(youth_unempl_rate, na.rm = TRUE),
            n = n(), .groups = "drop") %>%
  arrange(continent, income_status)
```

```
# Pivot wider for an easier comparison
```

```
continent_unemp %>%
  pivot_wider(id_cols = continent, names_from = income_status, values_from = mean_unemp)
```

```
## # A tibble: 6 x 3
##   continent      L      H
##   <chr>      <dbl> <dbl>
## 1 Africa     12.6  11.6
## 2 Americas   NA    19.9
## 3 Asia       26.0  11.7
## 4 Europe     NA    15.4
## 5 Oceania    NA    25.7
## 6 <NA>       NA    NaN
```

Simpson’s Paradox occurs when a trend appears in different groups of data but disappears or reverses when these groups are combined. When we look at the overall youth unemployment rate without grouping by continent, low-income (L) countries often report lower average youth unemployment compared to high-income (H) countries. This is counter-intuitive, but it stems from the fact that in low-income nations, a vast majority of the youth must work in the informal sector or agriculture out of sheer necessity, leaving fewer people “officially” unemployed.

Policy and institutional differences: Even within the same income classification, countries have different education systems, labor market regulations, and social safety nets that affect youth employment.

However, when we control for the confounding variable—the **continent**—the trend reverses or shifts. Within specific continents, high-income countries might actually exhibit lower youth unemployment than their poorer neighbors. The overall aggregation is heavily weighted by continents like Africa that consist of many low-income countries with low *reported* formal unemployment, thus skewing the global average and causing the paradox. By disaggregating the data by continent, we gain a more accurate view of the relationship between national income and youth unemployment within comparable regional contexts.

i. Data Export

Export the final tidy dataset from e. as a **CSV** with ; as a separator; . representing missing values; no row names included. Upload the .csv to TUWEL, together with the submission.

Answer

```
# Export the final tidy dataset
write_delim(df_vars, "df_vars_tidy.csv", delim = ";", na = ".")
```